

Sequence processing

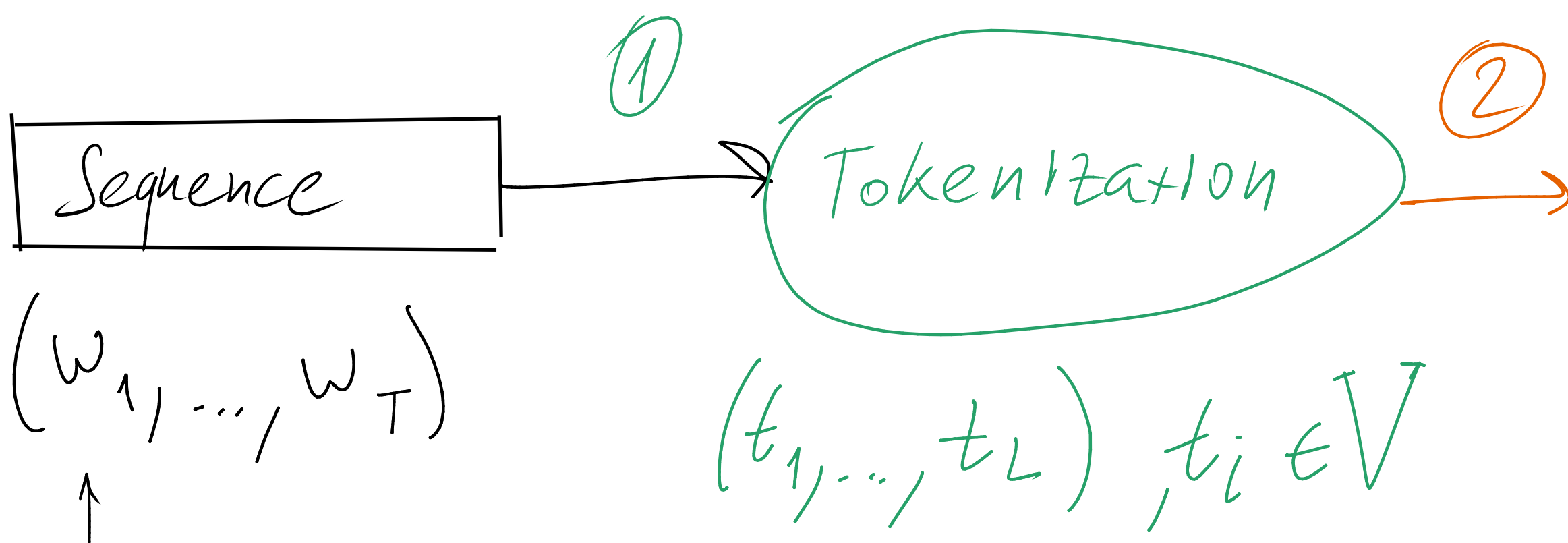
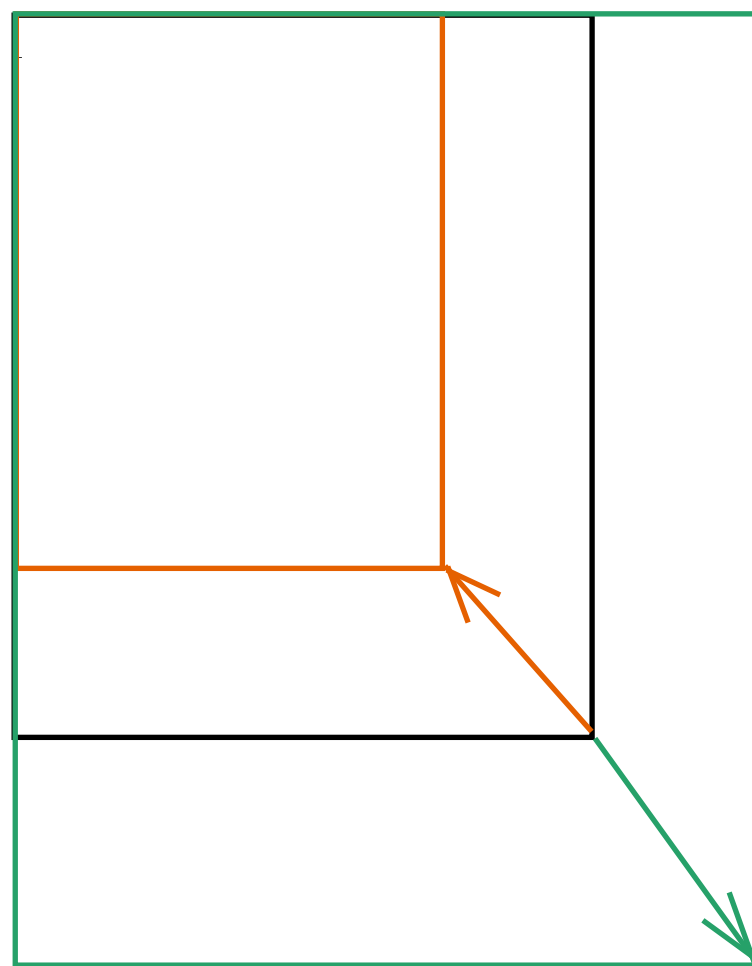
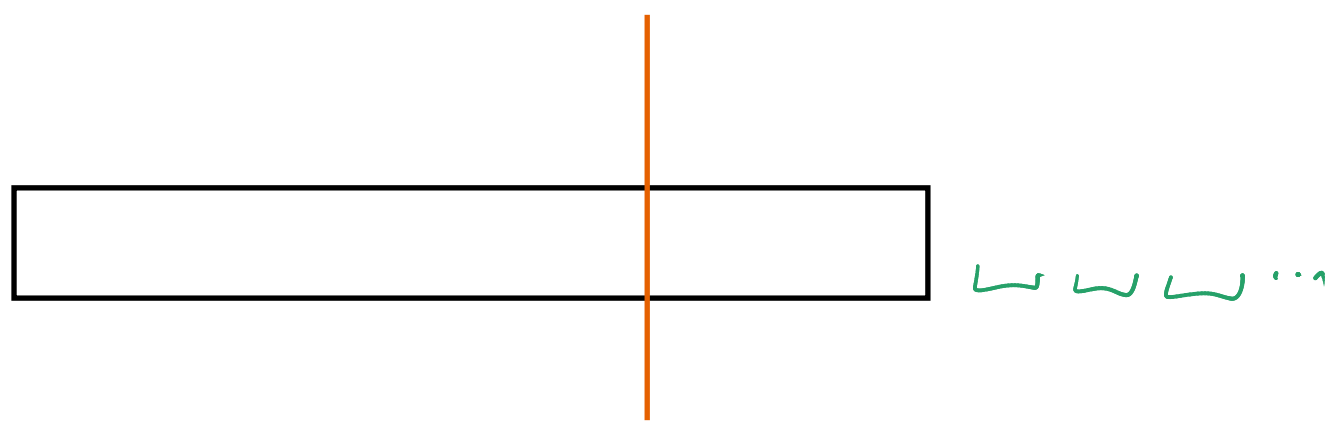
Processing Pipeline

- Text is discrete

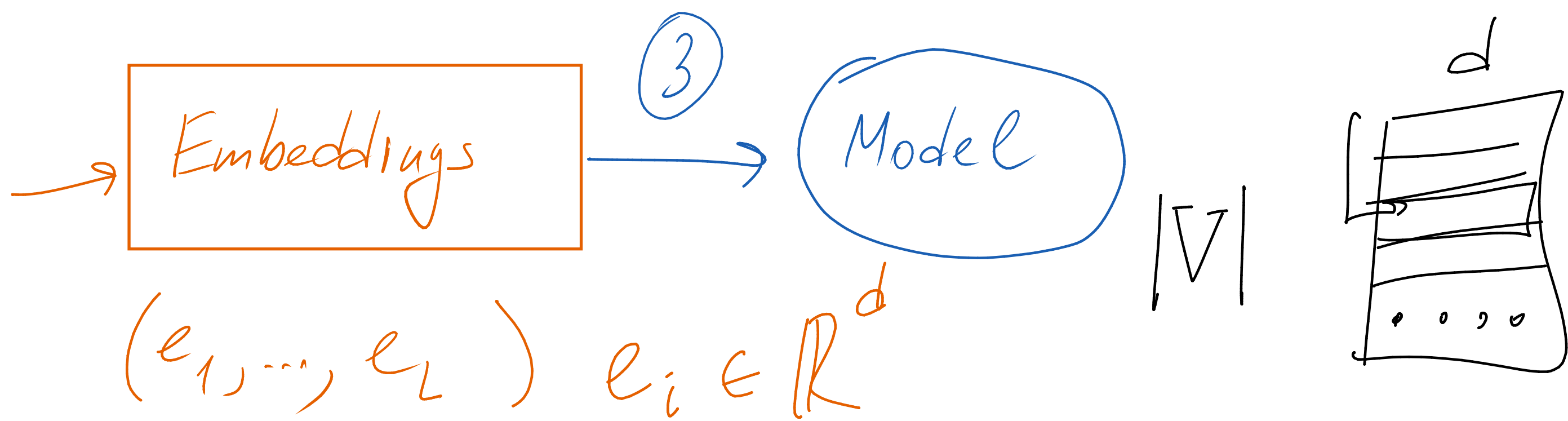
1 2 3 4

10010110 → 0 - 256

- Variable length

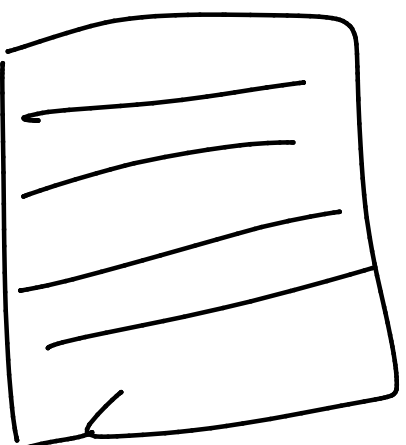


words



① • Vocabulary — set of tokens V

② • Embeddings e_i

③ L 
 d

Tokenization

На гворе права, на праве
грова.

• Character-level

|Н|а| |г|в|о|р|е| |т|р|а|в|а|,| |н|а| |т|р|а|в|е|
|г|р|о|в|а|.|

+ Very small vocab

+ Easy to encode new words

— Loses language semantics

— Very long sequences

• Word-level

| H a | г о в о р е | т р а в а | , | н а | т р а в е |
г о в о р а | . |

+ Preserve language semantics

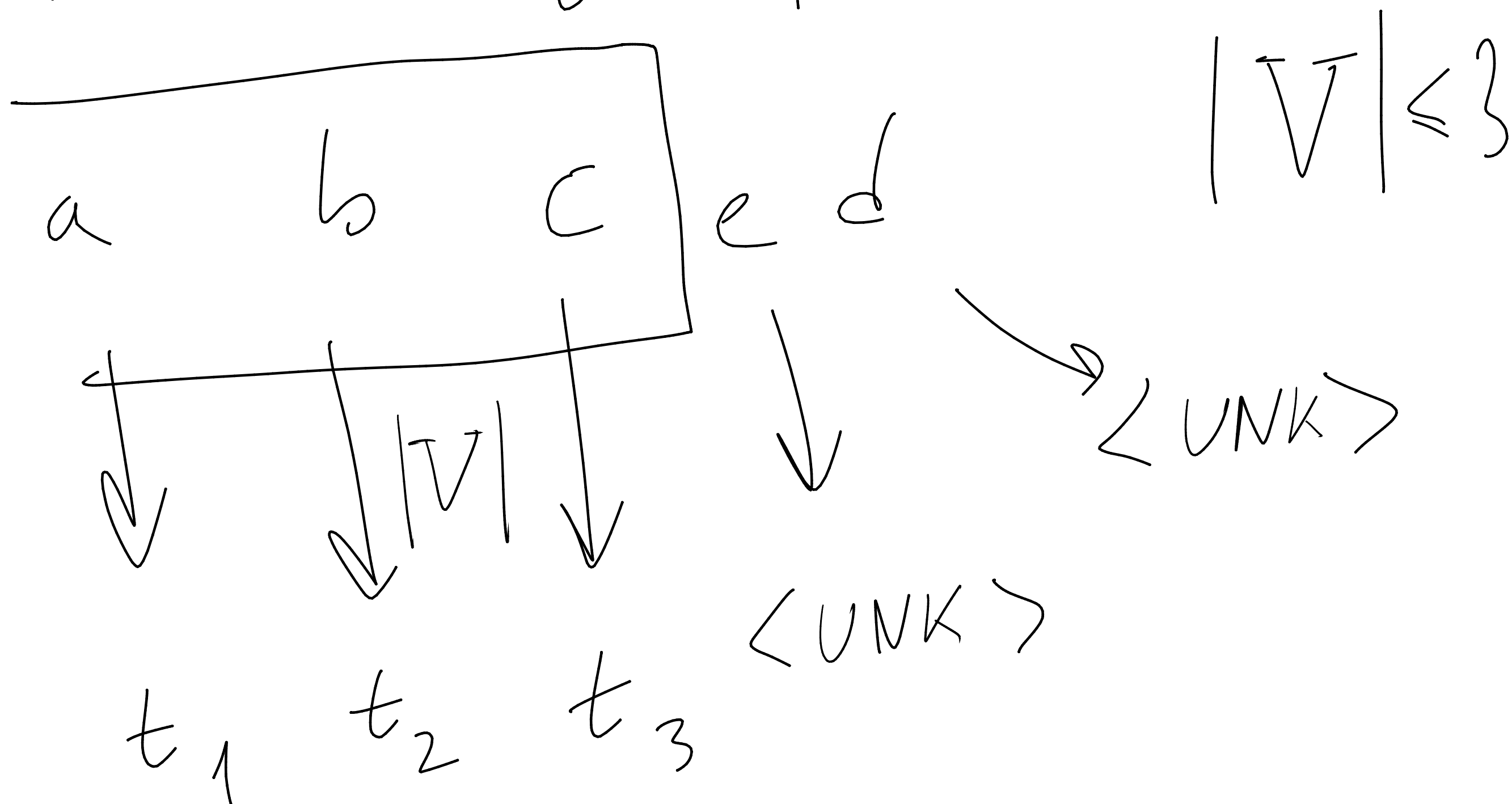
+ Shorter sequences

— Very large vocab

— No tokens for new words

• $\langle \text{UNK} \rangle$ — token for unknown words

a	b	c	d	e
32	16	5	1	3



- BPE Byte-pair encoding

1) Start with char-level

g H, a, g, b, o, p, e, 3, 1, . }

2) Find the most common n-grams,

add to V :

2-gram

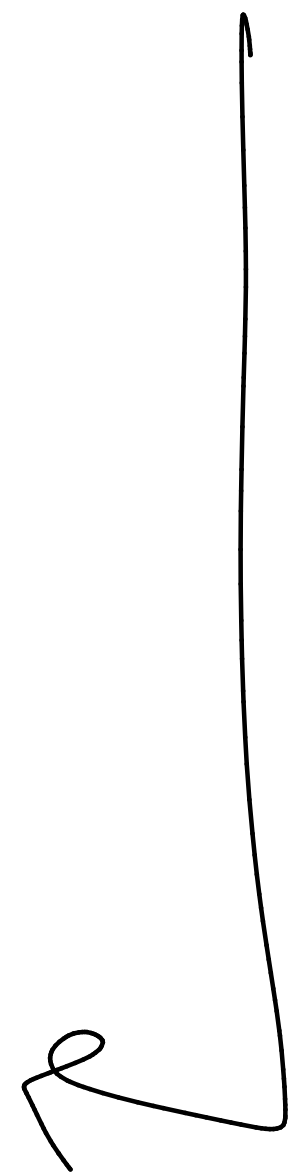
$\{H, a, g, b, o, p, e, \textcircled{3}, \textcircled{1}, \textcircled{1}, \textcircled{1}, \textcircled{1}, Ha, tp, pa, ab, ba\}$

3) 3-gram:

....., tpa, pab

4) $tpab$

While : $|V| \leq |V|_{\max}$

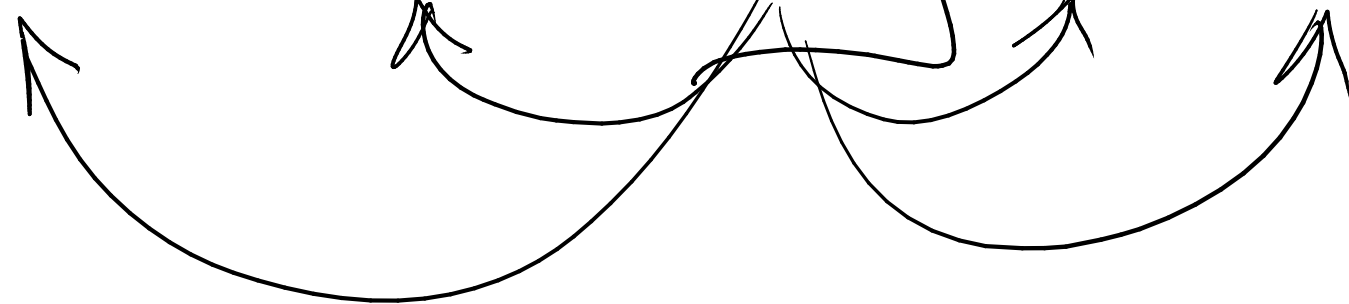


Embedding

Word2Vec

w_1 $\left[w_2 \quad w_3 \quad w_4 \quad w_5 \quad w_6 \right] w_7$

$C=5$



- Continuous bag of words (CBOW)

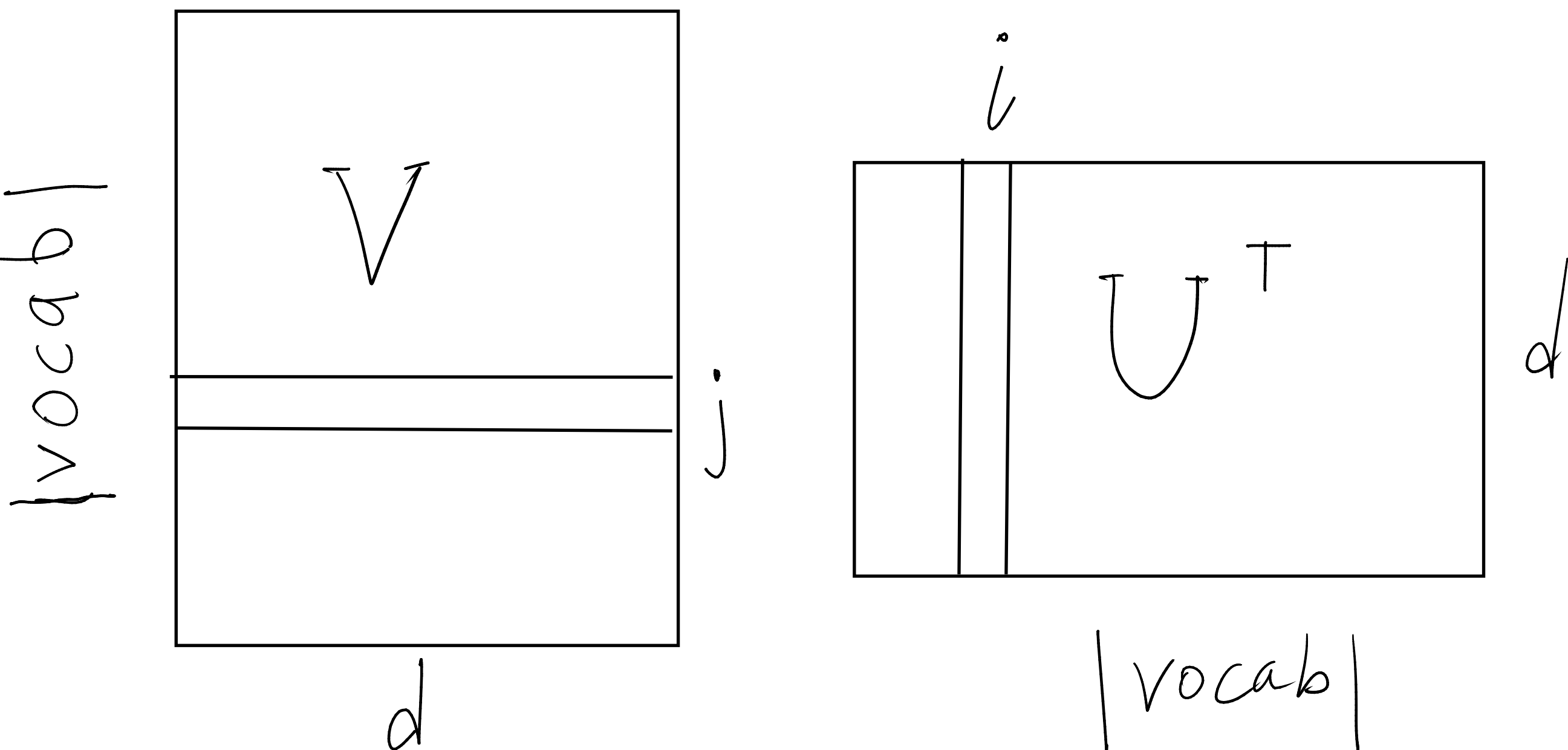
$$p(w_4 | w_2, w_3, w_5, w_6)$$

- Skip-gram

$$p(w_2, w_3, w_5, w_6 | w_4)$$

$$= p(w_2 | w_4) \cdot p(w_3 | w_4) \cdot \dots$$

assume independence



$$V_j \in \mathbb{R}^d \quad U_i \in \mathbb{R}^d$$

$$P(w_i | w_j) \approx \frac{\exp(V_j^T U_i)}{\sum_{k=1}^{|vocabulary|} \exp(V_j^T U_k)}$$

softmax

$$-\sum_{\substack{w_j - \text{center} \\ w_i - \text{context}}} \log P(w_i | w_j) \rightarrow \min_{V, U}$$

For CBOW

$$P(w_j | w_{j-H}, w_{j-H+1}, \dots, w_{j-1}, w_{j+1}, \dots, w_{j+H}) \\ = (V_{j-H} + V_{j-H+1} + \dots + V_{j+H})^T U_j$$

Classification:

Skip-gram w_i - input w_j - true class
 $\in \text{vocab}$

w_i - context
 w_j - center } "Do these words appear together"

w_i, w_j - input $0, 1$ - true class

$$p(w_i \text{ from context of } w_j) = \sigma(v_j^T u_i)$$

$$-\sum_{w_j \text{ - center}} \log(\sigma(v_j^T u_i)) \rightarrow \min_{V, U}$$

w_j - center

w_i - context

$$-\log(\underset{\uparrow}{\sigma}) - \log(1 - \sigma) \uparrow$$

$$-\sum \log \sigma(V_j^T u_i) - \sum \log(1 - \sigma(V_j^T u_i)) \rightarrow$$

w_j - center

w_i - context

w_j, w_i - negative pair

$$\rightarrow \min_{V, U}$$

Negative sampling

Final embedding:

- V

- U

- $\frac{1}{2}(V + U)$

CBOW - frequent tokens better

Skip-gram - rare tokens better